Differential Equations

Section 1: Separation of Variables

1 Solve the differential equation $\frac{dy}{dx} = \frac{2x}{y}$ with initial condition $y(0) = 4$ for $y > 0$.

a. $y = \sqrt{2x^2 + 16}$

b. $y = 2x^2 + 4$

c. $y = \sqrt{x^2 + 16}$

d. $y = x^2 + 4$

2 Solve $\frac{dy}{dx} = -3y$ with $y(0) = 50$.

a. $y = 50e^{-3x}$

b. $y = 50e^{3x}$

c. $y = -3x + 50$

d. $y = 50 - 3e^x$

3 Solve $\frac{dy}{dx} = (1 + y^2)e^x$ with $y(0) = 0$.

a. $y = \tan(e^x - 1)$

b. $y = \arctan(e^x - 1)$

c. $y = \tan(e^x)$

d. $y = e^x - 1$

Section 2: Integrating Factor Method for First-Order Linear ODEs

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4 For the linear differential equation $\frac{dy}{dx} + P(x)y = Q(x)$, what is the integrating factor $I(x)$?

a. $I(x) = e^{\int P(x)dx}$

b. $I(x) = \int P(x)dx$

c. $I(x) = e^{-\int P(x)dx}$

d. $I(x) = e^{\int Q(x)dx}$

5 Find the integrating factor for $\frac{dy}{dx} + \frac{3}{x}y = x^2$ for $x > 0$.

a. x^3

b. $3\ln x$

c. e^{3x}

d. $\frac{1}{x^3}$

6 Solve $\frac{dy}{dx} + 2y = 4$ with $y(0) = 5$.

a. $y = 2 + 3e^{-2x}$

b. $y = 2 + 5e^{-2x}$

c. $y = 4 + e^{-2x}$

d. $y = 2 - 3e^{-2x}$

Section 3: Homogeneous Differential Equations (y = vx)

7 Which substitution transforms a homogeneous differential equation $\frac{dy}{dx} = f(y/x)$ into a separable equation?

a. $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

b. $y = v + x$

c. $v = xy$

d. $y = v^2x$

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8 Classify the differential equation $\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$.

a. Homogeneous of degree 0

b. First-order linear

c. Exact non-linear

d. Second-order Bernoulli

9 Using $y = vx$, what does $\frac{dy}{dx} = \frac{y}{x} + 1$ become?

a. $x\frac{dv}{dx} = 1$

b. $v + \frac{dv}{dx} = 1$

c. $x\frac{dv}{dx} = v + 1$

d. $\frac{dv}{dx} = x$

Section 4: Numerical Euler's Method

10 What is Euler's recurrence relation for approximating solutions to $\frac{dy}{dx} = f(x, y)$ with step size h?

a. $y_{n+1} = y_n + hf(x_n, y_n)$

b. $y_{n+1} = y_n + \frac{h}{2}f(x_n, y_n)$

c. $y_{n+1} = f(x_n + h, y_n)$

d. $y_{n+1} = y_n + h^2 f(x_n, y_n)$

11 Given $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's method with step size $h = 0.1$ to find y_1 at $x = 0.1$.

a. 1.10

b. 1.05

c. 1.21

d. 1.15

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12 Using the result above ($x_1 = 0.1, y_1 = 1.10$), find y_2 at $x = 0.2$ with $h = 0.1$.

a. 1.22

b. 1.20

c. 1.25

d. 1.31

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